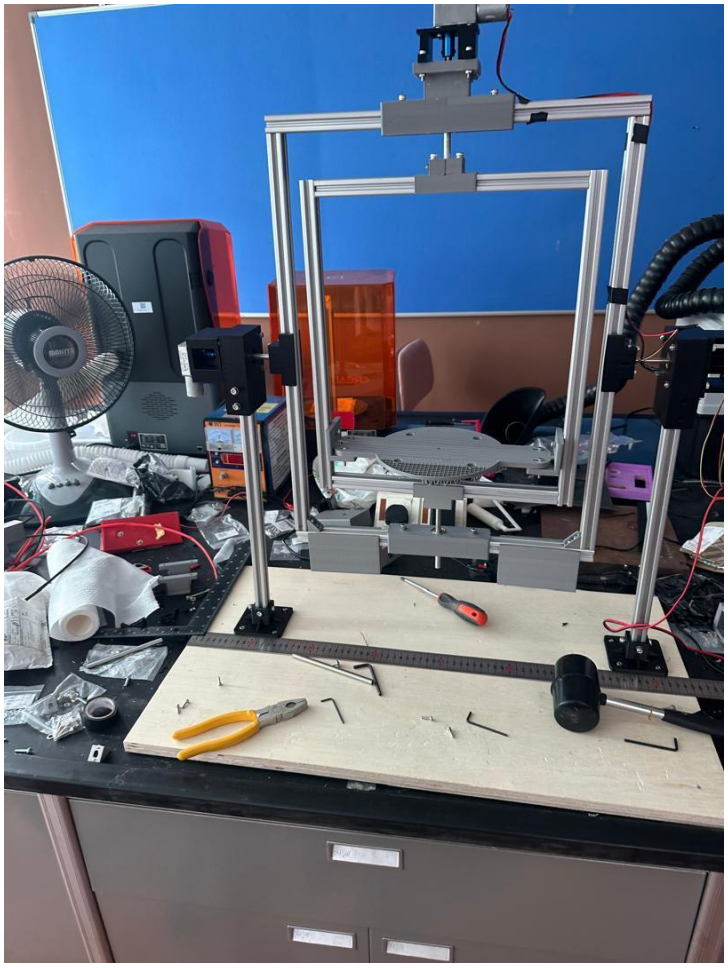
TECHNICAL PROJECT REPORT

Dual-Axis Silicone Mould Rotating Device

Design, analysis, fabrication, control and commissioning of a dual-axis rotational casting machine (rotocaster) for silicone moulds



Report number	TR-BFL-ROT-01, Revision 1.0
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Date	18 October 2026
Related documents	SOP-BFL-ROT-01 Standard Operating Procedure (written and visual)

Executive Summary

The Biofluid Laboratory needs hollow silicone models of organs and vessels whose closed shapes cannot be coated evenly by rotation about a single axis. Dr. Ahmad Zahran Md Khudzari therefore issued the project "Dual-Axis Silicone Mould Rotating Device": a device that turns a mould holder through 360° about two perpendicular axes to prevent silicone dripping, with two independent motor-driven shafts, a control panel and an LED status display. The brief set three tasks (the device, comprehensive technical documentation and an SOP), eight PDS items (a to h) and professional notes on documentation, quality control, safety and materials selection.

A dual-axis rotocaster was designed and built as a nested-box gimbal of 2020 aluminium extrusion that keeps a Ø300 × 300 mm mould at the intersection of its two perpendicular axes. Each axis is driven independently through 360° by a 12 V self-locking JGY-370 worm gearmotor coupled directly to an 8 mm shaft. The inner-axis motor is fed through a six-wire slip ring whose wires run through a hollow trunnion, so the cable cannot wind up. The whole machine was generated from one parametric CadQuery model, which also exported about 2 kg of 3D-printed parts. An Arduino Uno with an L298N driver, two speed potentiometers, a RUN/STOP switch and a 1.3-inch OLED display, housed in a custom 3D-printed box, controls the machine with a 15 s soft start. The mechanical build cost about RM 464, plus about RM 50 of electronics.

A torque and balance analysis based on the CAD solids showed that the machine is limited by unbalance rather than motor power: a 409 g counterweight is needed to balance the inner-axis motor, and with a 10 kg mould a residual offset of only 10 mm demands about 1 N·m, close to the 1.27 N·m rating of the fitted motors. The analysis also showed that the PDS target of 100 rpm cannot be met by a worm gearmotor at this scale and is unsuitable for the process, because above about 77 rpm the silicone is pinned to the wall and rotational casting becomes centrifugal casting. A load chart (5 kg with the fitted motors, 10 kg at up to 20 rpm with a 70 kg·cm motor) is recommended instead of a single maximum speed. Nineteen design and test faults were logged and corrected, including a motor that burnt out when it stalled against an over-heavy counterweight. The machine was balanced and run with independent speed control of both axes. A written and visual SOP was prepared.

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1 Introduction

1.1 Background

Rotational casting, or rotocasting, produces hollow parts by turning a closed mould that holds a measured amount of liquid material until the material gels on the inner wall. It needs no core, and the wall thickness is set by the amount of material and the motion. For a tube, rotation about its own axis is enough. For a closed, three-dimensional shape such as an aneurysm, a heart chamber or an organ, the liquid must be carried over every part of the wall, which needs rotation about two perpendicular axes. Rotational casting machines therefore turn the mould slowly about two axes at once, typically at 10–15 rpm, so that the liquid tumbles over the wall rather than being thrown against it.

1.2 Need

The laboratory's existing device turns a mould about one axis only. Silicone models with closed shapes could not be cast with an even wall. The project brief states the objective as developing "a sophisticated device capable of rotating a mould holder through dual-axis perpendicular orientations, enabling 360-degree rotation in all directions to prevent silicone dripping", with two independent motor-driven shafts, a comprehensive control panel and an LED status display for operational monitoring.

1.3 Objectives

(1) To design a dual-axis rotational casting machine that turns a Ø300 × 300 mm mould of up to 10 kg through 360° about two perpendicular axes.

(2) To size the motors, counterweight, bearings and shafts with a torque and balance analysis based on the CAD model.

(3) To build, balance and commission the machine at low cost using 3D-printed parts and off-the-shelf components.

(4) To develop an Arduino-based controller with independent speed control of each axis, a run switch, a display and a 3D-printed control box.

(5) To document the machine with a detailed SOP in written and visual formats and with this technical report.

1.4 Scope and deliverables

Table 1 Project tasks and deliverables

No.	Task in the brief	Where it is delivered
1	Design and fabricate a dual-axis rotary mechanism for silicone mould handling	The rotocaster in the Biofluid Laboratory; Sections 3 to 8
2	Prepare comprehensive technical documentation	This report, TR-BFL-ROT-01, with the CAD model, fault log, bill of materials, fastener schedule and control sketch
3	Develop Standard Operating Procedures (SOP) for device operation	SOP-BFL-ROT-01 with the visual SOP poster in its Appendix A; Section 9

2 Requirements

2.1 Product design specification

Table 2 lists the eight PDS items of the brief, items a to h, as written, with how each was interpreted. Two further requirements, i and j, were agreed with the supervisor at the start of the project.

Table 2 Product design specification of the dual-axis device

ID	PDS requirement (from the brief)	Interpretation in the design
a	Implement dual-axis configuration ensuring 360-degree rotational capability across orthogonal axes	Two perpendicular axes, each turning continuously through 360°, driven independently
b	Achieve smooth rotational operation at speeds up to 100 RPM	Smooth start and steady running; speed range to be checked against the drive and the casting process (Section 6.4)
c	Provide precise speed control mechanisms	Separate, repeatable speed setting for each axis with a soft start
d	Integrate wall-socketted power supply	Runs from a standard wall socket through a DC adapter
e	Install ON/OFF control functionality via operator panel	Run/stop control on a panel next to the speed controls
f	Incorporate LED display system for RPM and operational status monitoring	Display of the speed of each axis and the run state
g	Utilize 3D-printed or equivalent materials for component fabrication	Printed parts on the laboratory printers, combined with standard extrusion, shafts and bearings
h	Design mould holder with quick-release or minimal-effort mounting system	Standard bolt-on mould interface that any mould can be designed to
i	Mould envelope (agreed)	Ø300 × 300 mm, mass 2–10 kg
j	Cables (agreed)	Cables must not wind up during continuous rotation

2.2 Professional notes

The brief adds three professional notes. How each is met is set out in Section 10.2 and Section 10.3.

- **All tasks require detailed technical documentation.**
- **Emphasis on quality control and safety protocols.**
- **Materials selection based on mechanical properties and manufacturing feasibility.**

2.3 Constraints

- Small budget; parts available from online suppliers in Malaysia.
- Printed parts limited to the laboratory's Bambu Lab A1 (256 mm bed) and a smaller resin printer.
- Bench-top machine for a shared laboratory, operable by students after training.

3 Design Methodology

The project followed the process in Figure 1. The design was modelled before any part was made, checked by calculation and by automated geometry checks, and then built and tested in stages. Whenever a fault was found in printing, assembly or testing, the model was corrected and the affected parts were regenerated from it rather than modified by hand, so that the CAD files always matched the hardware. Every fault was recorded in a fault log (Section 8.3).

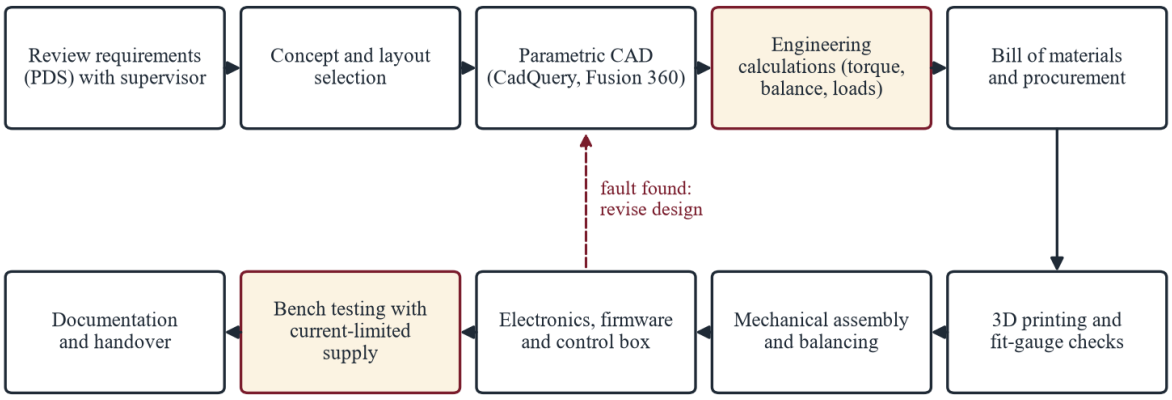


Figure 1 Development process used for the project.

The project ran over the twelve weeks of the training period alongside the single-axis upgrade (Figure 2). The aluminium extrusion was ordered first because it had the longest lead time, and the design was frozen in Week 8 with a planned budget of RM 367.

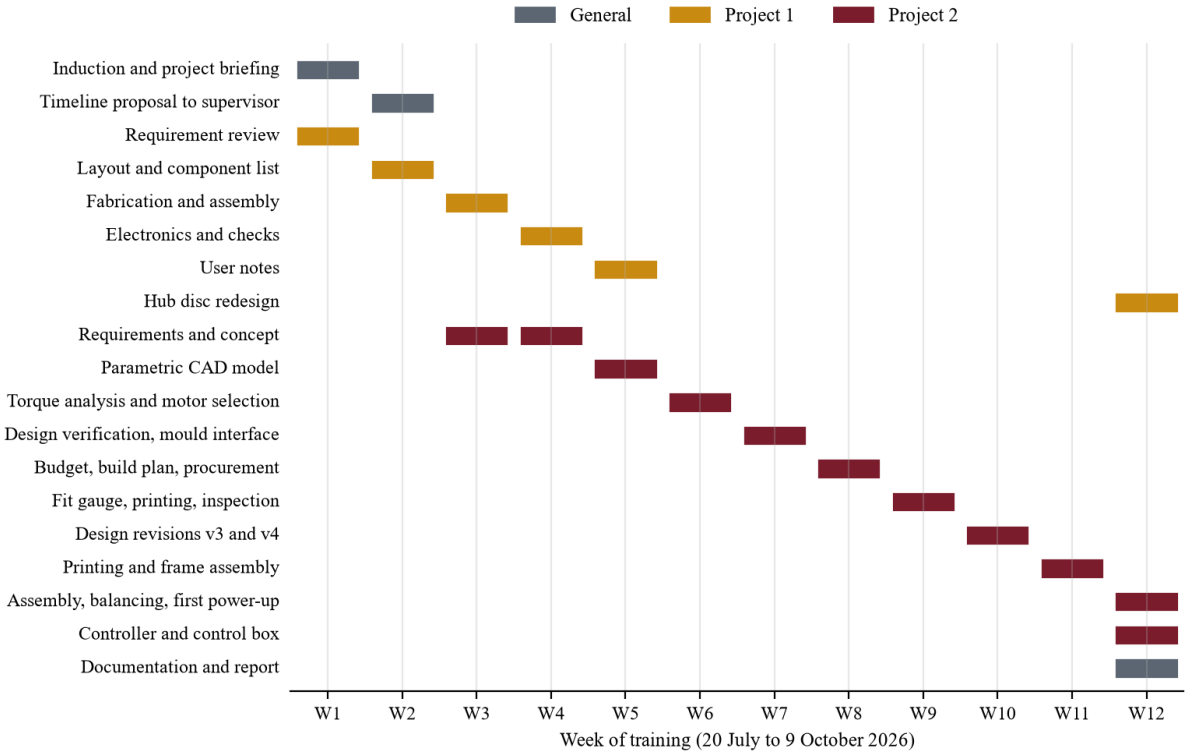


Figure 2 Project schedule (20 July to 9 October 2026); Project 2 is the dual-axis device.

4 Concept Selection

Three common layouts for a two-axis casting machine were considered (Table 3). The deciding factor was balance: because the motors are small worm gearmotors, every millimetre of offset between the mould and the axes becomes a torque that the motor must lift on every turn.

Table 3 Comparison of two-axis machine layouts

Layout	Advantages	Disadvantages	Decision
Nested-box gimbal (outer frame on trunnions, inner frame inside it)	Mould sits at the intersection of both axes, so unbalance is small; symmetric; full 360° on both axes; buildable from standard 2020 extrusion	Inner-axis motor rides on the outer frame and needs a slip ring; it also needs a counterweight	Selected
Offset arm (mould on an arm turned about a second axis)	Simple frame; mould easy to reach	Mould far from the main axis gives a large unbalance torque and needs heavy counterweights	Rejected
Rock-and-roll (full turn on one axis, rocking on the other)	Simple drive; no slip ring	Second axis only rocks, so closed shapes are not fully coated	Rejected

5 Detailed Design

5.1 System architecture

Figure 3 shows how power, control and drive are connected. A 12 V adapter feeds the Arduino and the L298N driver. The Arduino reads the two speed potentiometers and the RUN/STOP switch and drives both motors through the L298N, showing the state on the OLED. Motor 1 turns the outer frame directly; Motor 2 rides on the outer frame, is fed through the slip ring and turns the inner frame, which carries the mould.

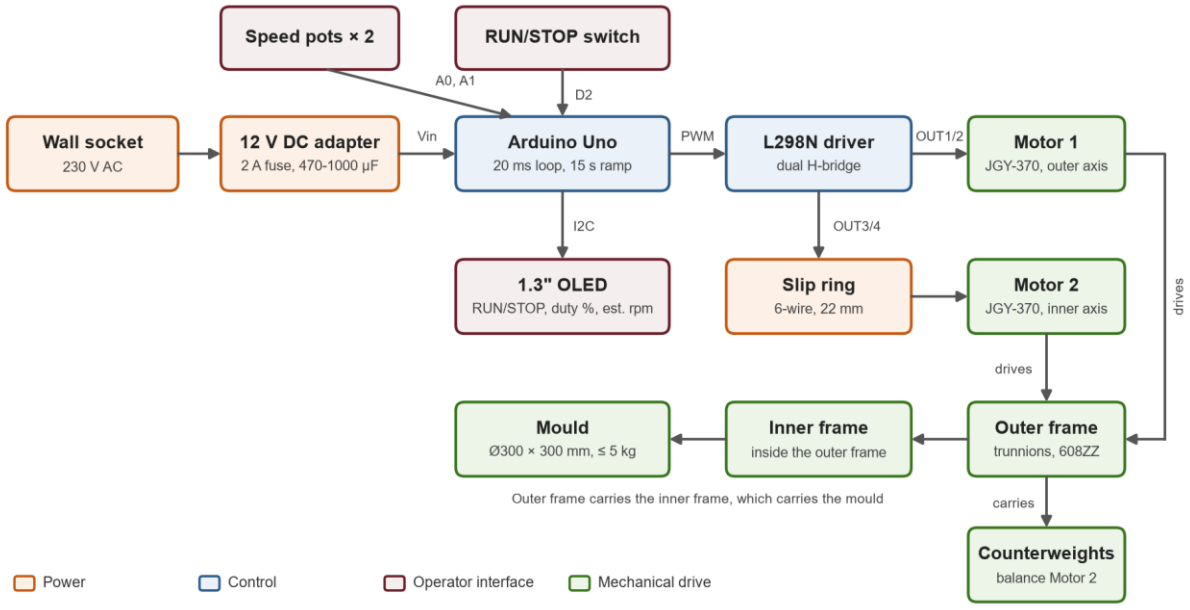
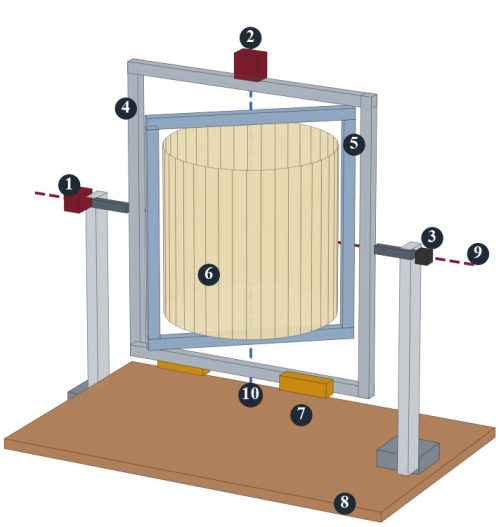


Figure 3 System block diagram of the dual-axis rotating device.

5.2 Mechanical layout

The outer frame turns about a horizontal axis on two trunnion shafts carried by vertical tower posts. The inner frame turns inside it about the perpendicular axis, and the mould is bolted to a plate on the inner frame so that it sits where the two axes cross (Figure 4). Each axis has its own JGY-370 worm gearmotor, coupled directly to an 8 mm shaft through a rigid 6–8 mm coupler, so there are no belts or gears to adjust. The worm gearbox is self-locking, which holds the frames in place when the power is off. The shafts run in 608ZZ bearings seated in printed housings. The machine stands on an 18 mm plywood base of 750 × 450 mm, and the posts drop into printed sockets that are screwed to the base and locked with M5 side screws. The main specifications are listed in Table 4.



Key

1 Motor 1, outer axis (JGY-370)

2 Motor 2, inner axis

3 Slip ring (6-wire, 22 mm)

4 Outer frame, 468 x 536 mm

5 Inner frame, 360 x 416 mm

6 Mould envelope, Ø300 x 300 mm

7 Counterweights (M8-nut ballast)

8 Plywood base, 750 x 450 x 18 mm

9 Outer (horizontal) axis, 375.8 mm high

10 Inner (vertical) axis

Figure 4 Simplified schematic of the dual-axis rotocaster.

Table 4 Main specifications of the rotocaster

Item	Specification
Layout	Nested-box gimbal; mould at the intersection of the two axes
Frame	2020 aluminium extrusion (6 mm slot) with gusset corner brackets
Extrusion cut list	4 × 184, 2 × 536, 2 × 320, 2 × 416 and 2 × 351.8 mm (3.98 m in total)
Outer frame	468 × 536 mm as built; horizontal axis 375.8 mm above the base
Inner frame	360 × 416 mm; clear opening Ø320 mm
Mould envelope	Ø300 × 300 mm (up to 380 mm high), 2–10 kg
Mould interface	Base: Ø140 mm PCD, 4 × M6. Rim: Ø280 mm PCD, 6 × M5. Plate slotted ±6 mm
Drives	2 × JGY-370 4632 worm gearmotor, 12 V, 18 rpm (13 rpm loaded), 13 kg·cm, self-locking
Couplings and shafts	Rigid 6–8 mm couplers; Ø8 mm hardened steel shafts with 1 mm deep D-flats; right trunnion Ø8 mm tube with 3 mm bore
Bearings	608ZZ (8 × 22 × 7 mm) in printed bores of 22.2 mm
Power transfer	6-wire, 22 mm capsule slip ring, 2 A per circuit
Counterweight	2 holders of M8-nut ballast on the bottom rail at x = ±120 mm
Base	Plywood 750 × 450 × 18 mm with printed post sockets
Rotating mass	About 5.35 kg without mould

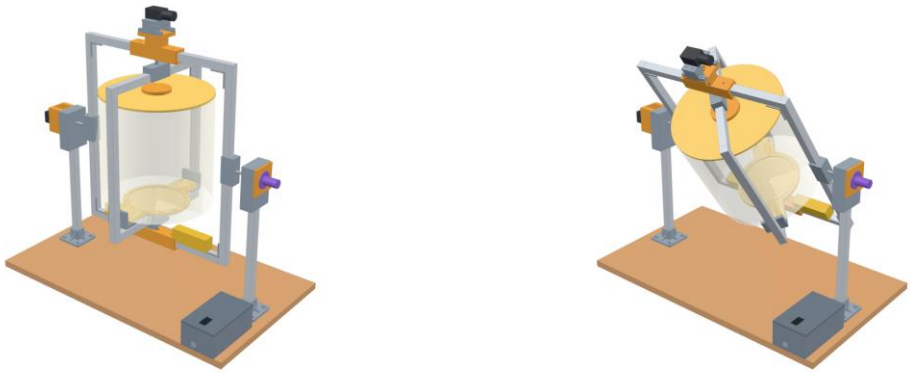
5.3 Parametric CAD model

The machine was modelled in CadQuery, a Python library for parametric CAD. One script generates every part from a single set of parameters (Table 5) and exports the printed parts as STL files; supporting modules generate the connector plate, mould lid, counterweight and slip-ring parts. When a delivered part differed from the design, only a parameter had to change. For example, the 180 mm rails were delivered at 184 mm, and raising the inner-to-outer frame clearance from 30.0 to 33.35 mm absorbed the difference without changing the shape of any printed part. A simplified three-body assembly with two revolute joints was built in Fusion 360 to confirm that both frames turn through a full 360° without interference.

Table 5 Main parameters of the parametric CAD model

Parameter	Value	Reason
Mould diameter × height	300 × 300 mm	Set by the supervisor
Extrusion size and slot	20 mm, 6.2 mm slot	2020 profile
Socket clearance per side	0.35 mm	Printed sockets slide onto the rail
Printed wall around a socket	5.0 mm	Stiffness of the brackets
Rail depth in a socket	45 mm	Clamping length
Inner frame clearance (Z)	18 mm	Inner frame clears the outer frame
Outer frame clearance (X)	33.35 mm	Absorbs the 184 mm rails
Outer frame clearance (Z)	24 mm	Sweep of the inner frame
Ground clearance	38 mm	Outer frame clears the base
Shaft diameter	8 mm	608 bearing bore
Bearing bore × width	22.2 × 7 mm	From the bearing fit gauge (part 00)
Gearbox size	46 × 32 × 25.2 mm	JGY-370 4632 drawing
Motor hole pattern	18 × 33 mm, Ø3.4 mm	Supplier drawing
Slip ring body	Ø22 × 40 mm	Capsule slip ring

The complete machine was also assembled as an interactive 3D model from the parametric dimensions and the STL files (Figures 5 and 6). In the web viewer the two axes can be turned with sliders and each part is named when the pointer is over it, so it can be used to explain the machine to new users.



(a) At rest (b) Outer frame at 35°, inner frame at 40°

Figure 5 3D model of the complete rotocaster: (a) at rest; (b) outer frame at 35°, inner frame at 40°.

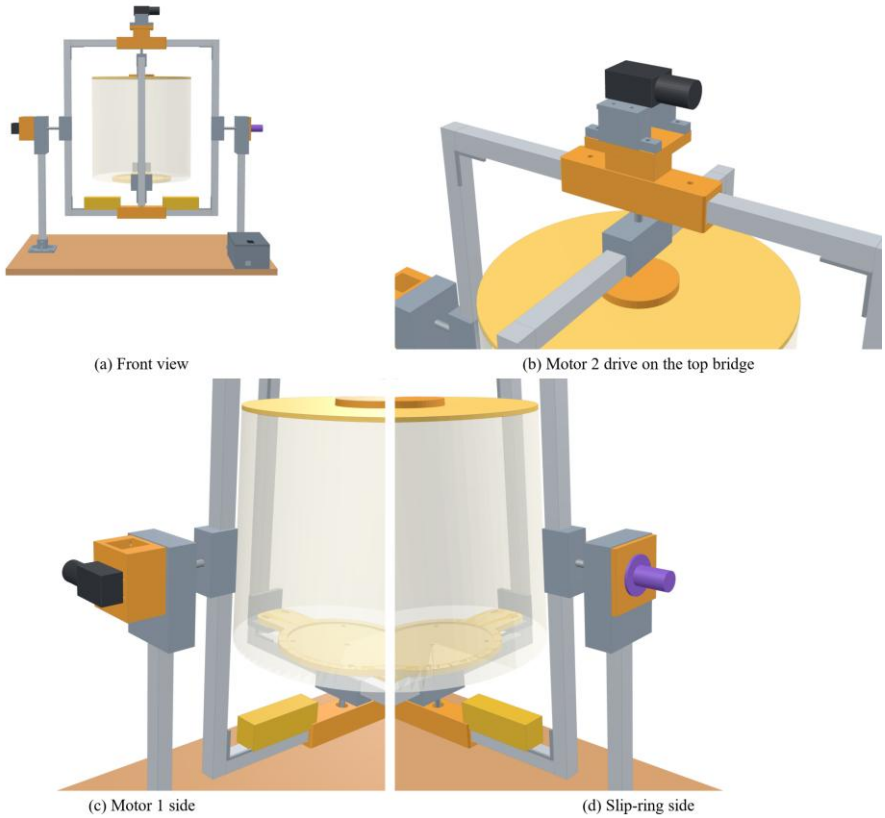
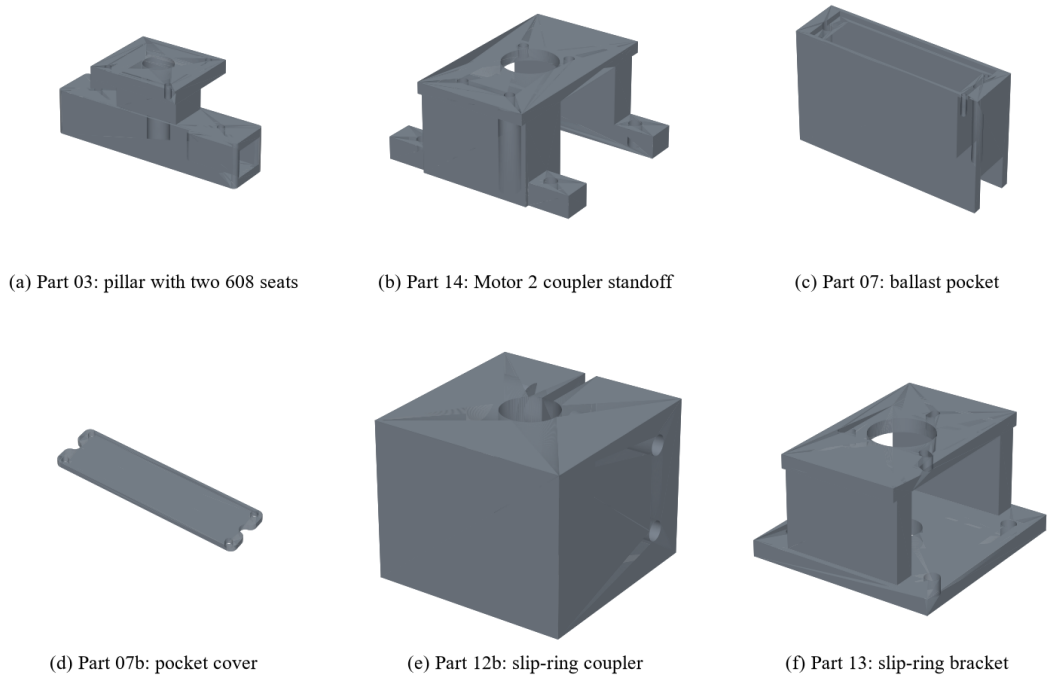


Figure 6 Detail views of the 3D model: Motor 2 drive on the top bridge, Motor 1 at the left tower and the slip ring at the right tower.

5.4 Inner-axis drive, counterweight and slip ring

Three areas needed particular attention (Figure 7). In the first design of the inner-axis drive, the stub shaft ended 17 mm short of the coupler, and the coupler had to sit in an 8 mm gap that could pass neither the shaft nor the 20 mm coupler and left the grub screws unreachable. In the final revision the top bridge (part 03) was rebuilt as a solid 50 × 30 × 28 mm pillar with the shaft bore running straight through and a second 608 bearing in its top plate, and a new standoff (part 14) carries Motor 2 above the plate with the coupler in an open 29 mm high space that gives access for a hex key. The shaft now runs on two bearings 54 mm apart, and its free end beyond the upper bearing dropped from 75 mm to 16 mm (Table 6).



(a) Part 03: pillar with two 608 seats (b) Part 14: Motor 2 coupler standoff (c) Part 07: ballast pocket

(d) Part 07b: pocket cover (e) Part 12b: slip-ring coupler (f) Part 13: slip-ring bracket

Figure 7 Motor 2 drive, counterweight and slip-ring parts.

Table 6 Inner-axis drive stack

Element	Height above the base (mm)
Inner stub shaft (125 mm long)	200 to 325
Shaft gripper bore	208 to 224
Lower 608 bearing	241 to 248
Upper 608 bearing	302 to 309
Top face of the part 03 plate	309
Coupler space (29 mm tall, 23 mm wide)	309 to 338
Motor 2 mounting face	342

The counterweight pocket was first modelled as a sealed void that could never be filled. It was reopened as an 82 × 22 × 39 mm pocket (70.4 cm³) that holds about 304 g of M8 nuts against the 204 g needed per side, and closed by a screwed cover (part 07b), because the weight turns upside down on every revolution. The slip ring's flange is held on the right tower by a bracket (part 13) with radial slots, so the ring can be rotated until its holes line up, and a small coupler (part 12b) joins the hollow trunnion to the rotor stub with two pinch screws. The right trunnion is an 8 mm tube with a 3 mm bore, and the six rotor wires run through its centre to Motor 2. Because a 1 mm D-flat would cut the tube wall from 2.5 to 1.5 mm, that trunnion is held by a clamping gripper (part 02b).

5.5 Mould interface

So that future moulds can be designed without the machine at hand, a mould interface standard was defined and the matching parts were modelled (Figure 8). The mould foot bolts to the 180 × 300 × 8 mm mould plate (part 08) on a Ø140 mm pitch circle with four M6 screws; the plate rests on two shelves on the inner frame, and its slots allow ±6 mm of movement for balancing. The mould rim bolts to the Ø300 mm lid (part 10) on a Ø280 mm pitch circle with six M5 screws, tightened in a star pattern. The lid has a spigot ring underneath and a Ø45 mm pour port closed by a pour plug (part 11) with an O-ring and three M4 screws, with positive retention so that it cannot fall out when the mould is inverted. This meets the no-dripping requirement. The mould wall must stay within about Ø125 mm

at the base so that the foot screws clear it, and the rim is flush at Ø300 mm so that it never overhangs the Ø320 mm opening.



(a) Part 08 mould plate, 180 × 300 × 8 mm (b) Part 10 mould lid, Ø300 mm (c) Part 11 pour plug

Figure 8 Mould interface parts: (a) mould plate, (b) mould lid and (c) pour plug.

6 Engineering Analysis

6.1 Torque model

The torque required from each motor was derived from the CAD model rather than estimated. A motor driving a rotating frame must overcome four loads at the same time:

$$T_{req} = T_{unb} + T_{acc} + T_f + T_p \tag{1}$$

where T_{unb} is the torque needed to lift an off-centre mass, T_{acc} the torque to accelerate the assembly, T_f the bearing friction and T_p the torque to shear the liquid silicone. Centrifugal force acts radially on the bearings and produces no torque about either axis, so it was checked separately as a bearing load. Gyroscopic coupling between the axes is below 0.01 N·m at 10–20 rpm and was neglected.

6.2 Mass properties and unbalance

The masses and centres of gravity of all parts were found by volume integration of the CAD solids, using densities of 2700 kg/m³ for the aluminium extrusion, 744 kg/m³ for PLA printed with four walls and 40 % infill, and 7850 kg/m³ for the steel shafts (Table 7).

Table 7 Mass and centre of gravity of the rotating parts

Part	Mass (g)	CG height above outer axis (mm)
Outer frame (rails, hubs, trunnions)	2669	+24.4
Inner frame	2113	0.0
Connector plate	164	+3.4
Shelf brackets (2 off)	48	−151.2
Counterweight body (empty)	83	+35.8
Motor 2	162	+268.0
Coupler	30	+258.0
Total rotating mass, no mould	**5350**	

The height of the combined centre of gravity above the outer axis, and the permanent unbalance torque it produces, are:

$$z_G = \Sigma\, m_i\, z_i \,/\, \Sigma\, m_i \approx 20.5\ mm \tag{2}$$

$$T_{unb} = m\, g\, z_G = 5.35 \times 9.81 \times 0.0205 \approx 1.076\ N{\cdot}m \tag{3}$$

Motor 2 is the main cause: it weighs only 162 g but sits 268 mm from the axis. The counterweight needed on the bottom rail at the same radius is:

$$m_{cw} = T_{unb} \,/\, (g\, r_{cw}) = 1.076 \,/\, (9.81 \times 0.268) \approx 0.409\ kg \tag{4}$$

This 409 g of M8-nut ballast is split between two pockets of about 204 g at x = ±120 mm. Without it, 1.076 N·m of motor torque would be lost before any mould is fitted, which is 85 % of the motor rating.

6.3 Acceleration, friction and fluid torque

With a 10 kg mould modelled as a thin-walled Ø300 × 300 mm cylinder, the inertia about the outer axis rises from 0.2475 to 0.435 kg·m². Reaching 20 rpm (2.094 rad/s) over the 15 s ramp used in the controller needs:

$$T_{acc} = I\,\omega / t_{ramp} = 0.435 \times 2.094 / 15 \approx 0.061\ N{\cdot}m \tag{5}$$

Friction in the four 608 bearings is about 0.001 N·m, and shearing a 3 mm silicone film at 20 rpm takes less than 0.01 N·m. If the mould's centre of gravity is brought within 10 mm of the axis with the ±6 mm plate slots, the residual unbalance of a 10 kg mould is:

$$T_{res} = m_{mould}\,g\,e = 10 \times 9.81 \times 0.010 \approx 0.98\ N{\cdot}m \tag{6}$$

The total demand on the outer axis is given in Table 8. It hardly changes with speed: at 10 kg the residual unbalance alone is 0.98 N·m, while acceleration adds only 0.03–0.30 N·m. The machine is therefore **unbalance-limited rather than power-limited**, and careful balancing matters more than the motor rating.

Table 8 Torque demand on the outer axis in N·m (residual mould offset 10 mm)

Mould mass	10 rpm	20 rpm	30 rpm	100 rpm
2 kg	0.22	0.24	0.26	0.40
5 kg	0.51	0.54	0.56	0.73
10 kg	1.01	1.04	1.07	1.28

6.4 Motor selection and the speed requirement

The output torque of a worm gearmotor is set by its gear ratio, so faster versions deliver much less torque. For the JGY-370 family the rated torque falls about fifteen times between 10 and 100 rpm (Table 9 and Figure 9). At 100 rpm the gearbox delivers only 0.15 N·m, while the empty gimbal alone needs 0.20 N·m to accelerate in 15 s, so the 100 rpm requirement cannot be met by this class of motor.

Table 9 Comparison of candidate motors against the torque demand

Candidate motor	Torque (N·m)	2 kg	5 kg	10 kg	SF at 10 kg
JGY-370 at 10 rpm	2.21	Pass	Pass	Pass	2.2
JGY-370 at 20 rpm	1.18	Pass	Pass	Pass	1.1
JGY-370 at 30 rpm	0.73	Pass	Pass	Fail	0.7
JGY-370 at 100 rpm	0.15	Fail	Fail	Fail	0.1
Worm gearmotor, 16 rpm, 70 kg·cm	6.87	Pass	Pass	Pass	6.8

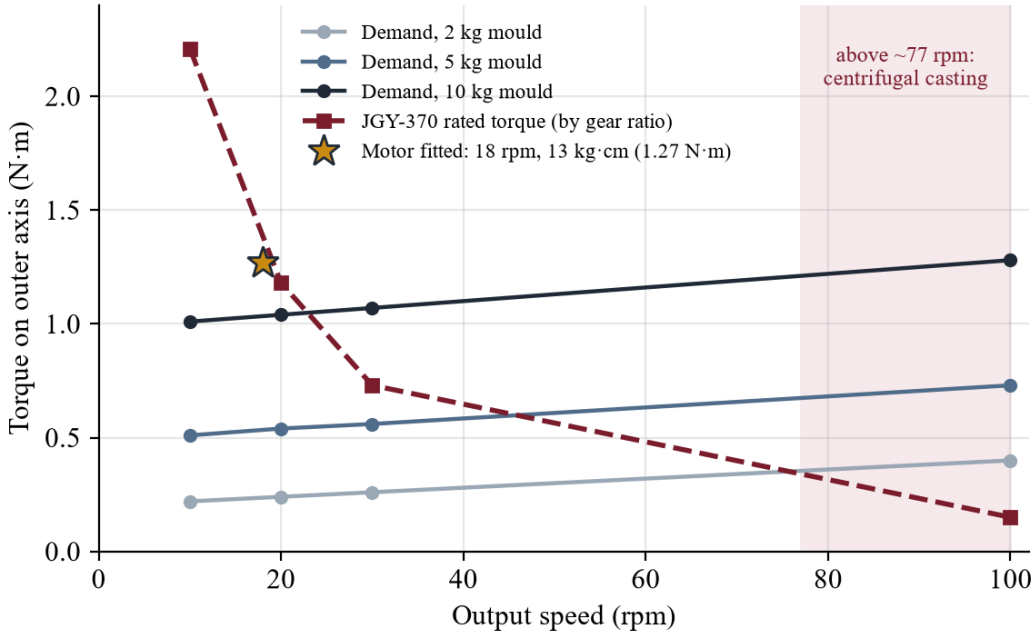


Figure 9 Torque demand on the outer axis compared with the worm gearmotor characteristic.

A speed of 100 rpm is also undesirable for the process. The speed at which the centrifugal acceleration at the mould wall (r = 0.15 m) equals gravity is:

$$n_c = (60 / 2\pi)\sqrt{(g / r)} = 9.549 \times \sqrt{(9.81 / 0.15)} \approx 77\ rpm \quad (7)$$

Above this speed the silicone is pinned to the outer wall instead of tumbling, and the process becomes centrifugal casting, which gives thick walls at the equator and starved poles. This finding was reported to the supervisor with a recommendation to rate the machine with a load chart, allowing the full 10 kg at 20 rpm or less, instead of a single maximum speed.

The calculation recommends a 16 rpm, 70 kg·cm (6.87 N·m) worm gearmotor, which gives a safety factor of 6.8 with a 10 kg mould. The 12 V JGY-370 4632 at 18 rpm (13 rpm loaded, 1.27 N·m, RM 27.44 each) had been purchased before the speed decision was confirmed. It meets the demand with safety factors of about 1.2, 2.4 and 5.7 for 10, 5 and 2 kg moulds respectively, so moulds are limited to 5 kg in the SOP unless the stronger motor is fitted. Motor 2 on the inner axis carries a much lower load: with the mould centred, its demand is mainly inertia, about 0.04 N·m for a 10 kg mould at 20 rpm.

6.5 Verification of the other load-bearing parts

Table 10 Verification of the other load-bearing components

Component	Load	Capacity	Margin
Trunnion bearing, each (10 kg mould)	75 N	3400 N (dynamic)	45 ×
D-flat bearing stress in PLA	2.8 MPa	about 45 MPa (yield)	16 ×
Slip ring current	0.35 A running	2 A per circuit	5.7 ×
Ø8 mm shaft bending, 150 mm span	12 MPa	250 MPa (yield)	21 ×

All have large margins, so the motor is the weakest link. This is the right place for it: a stalled motor stops the machine, whereas a failed bracket would drop the mould.

7 Electronics and Control

7.1 Hardware

The controller is based on an Arduino Uno and an L298N dual H-bridge module (Figure 10 and Table 11). Each motor has its own channel: the enable inputs ENA and ENB receive PWM from pins D5 and D6 to set the speed, and IN1–IN4 on D7, D8, D9 and D10 set the direction. Two potentiometers on A0 and A1 set the speed of the outer and inner axes, an ON-OFF-ON rocker switch on D2 acts as the RUN/STOP switch, and a 1.3-inch 128 × 64 I2C OLED on A4 and A5 shows the state of the machine. A 12 V adapter supplies the L298N and, through Vin, the Arduino, with a 2 A fuse in the positive line, and Motor 2 is fed through the slip ring. A 0.1 µF capacitor across each motor and a 470–1000 µF capacitor across the 12 V input suppress electrical noise.

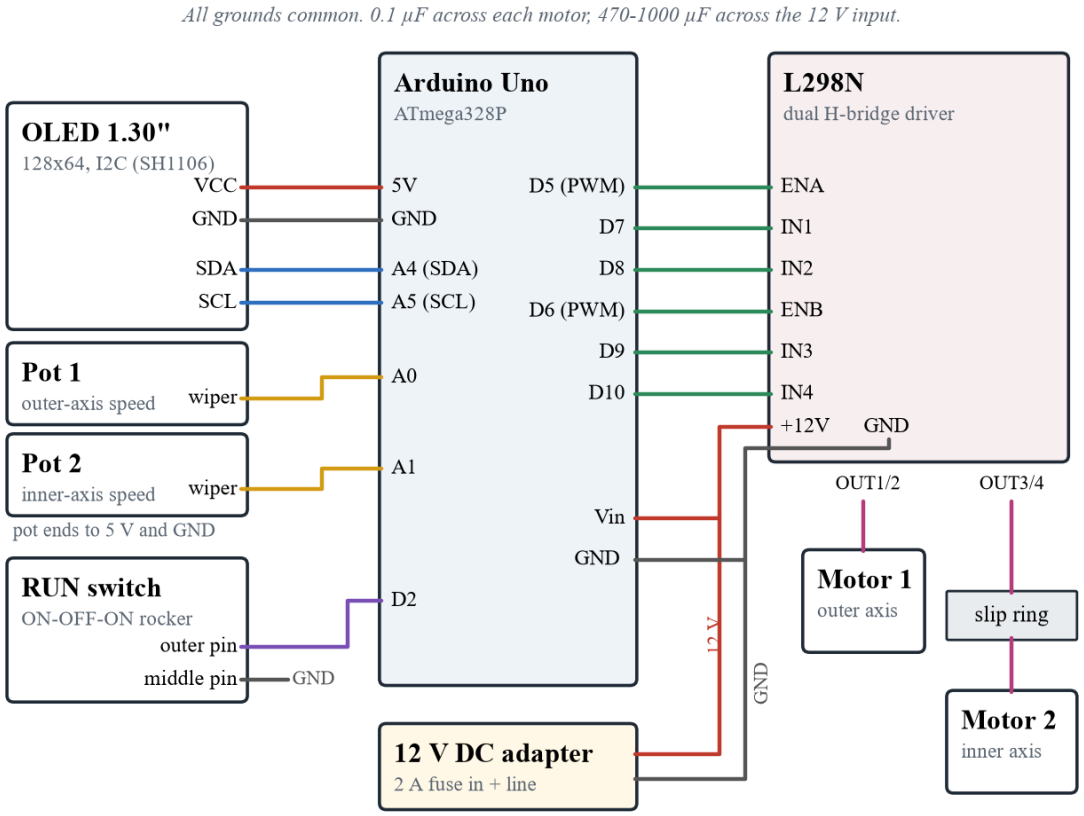


Figure 10 Wiring diagram of the rotocaster controller.

Table 11 Pin assignment of the controller

Arduino pin	Connected to	Function
D5 (PWM)	L298N ENA	Speed of Motor 1 (outer axis)
D7, D8	L298N IN1, IN2	Direction of Motor 1
D6 (PWM)	L298N ENB	Speed of Motor 2 (inner axis)
D9, D10	L298N IN3, IN4	Direction of Motor 2
A0	Potentiometer 1 wiper	Outer-axis speed setting
A1	Potentiometer 2 wiper	Inner-axis speed setting
D2	Rocker switch (to GND)	RUN/STOP input with internal pull-up
A4, A5	OLED SDA, SCL	I2C display
Vin, GND	12 V adapter via 2 A fuse	Power; all grounds common

7.2 Firmware

The control sketch (Appendix C) follows the flow in Figure 11 and runs a 20 ms loop. When the RUN switch is closed, it reads each potentiometer as the average of eight samples, maps the reading to a PWM value between 60 and 255 with a dead band near zero, and ramps the output towards that value so that full speed is reached in 15 s, the same ramp time used in the torque calculation. A slow ramp avoids sloshing the silicone and limits the starting current. A running motor is never held below the minimum PWM, so it cannot sit stalled in the low range. Opening the switch or turning a potentiometer to zero stops the motor at once, and the self-locking worm gear holds the frame. The OLED is refreshed every 300 ms with the RUN/STOP state and, for each axis, the duty cycle and an estimated speed. Because there is no encoder, the displayed speed is estimated from the duty cycle and the 18 rpm rating. The ramp uses the real elapsed time, so it stays at 15 s even while the display is being redrawn.

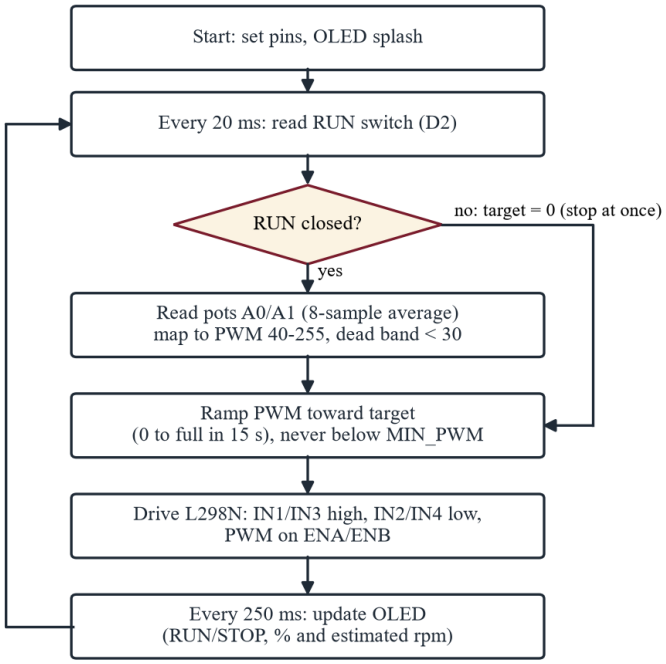


Figure 11 Flowchart of the control sketch.

7.3 Control box

A control box was designed in CadQuery to house all the electronics (Figure 12). Its 160 × 90 × 60 mm cavity leaves room for the wiring, and the overall size is 166 × 96 × 63 mm. The Arduino and the L298N sit on printed standoffs, with the L298N turned so that its power terminals face the cable wall. The left wall has an opening aligned with the Arduino USB port, so the board can be programmed without opening the box, and the right wall has a Ø10 mm entry for the adapter cable and a Ø16 mm exit for the motor cables. Vent slots let the L298N heat sink breathe. The 5 mm lid carries a window for the OLED, two potentiometer holes with Ø14 mm recesses so that the short pot bushings still reach their nuts, and a cut-out for the snap-in rocker switch, where the lid is thinned to 1.6 mm so the switch clips can grip. The labels OUTER, INNER and RUN and the project title are engraved on the lid. A collision check in the script confirmed that the boards, plugs, cables, potentiometers and switch clear the walls and the lid, and every screw length was checked against the material it passes through.



(a) Body: board standoffs, USB, power and cable holes (b) Lid, outer face: OLED window, pots and switch

Figure 12 CAD model of the 3D-printed control box: (a) body; (b) lid.

8 Fabrication, Assembly and Commissioning

8.1 3D printing

The printed parts (Figure 13 and Table 12) were printed in PLA at 0.2 mm layer height and 40 % infill, with six walls on parts that carry bearing press fits. Printed holes come out slightly undersize, so a bearing fit gauge (part 00) with a range of bore sizes was printed first, and the bearing bore in the model was set to 22.2 mm from its result. The mould plate and lid are larger than the print bed, so the model exports them already split: the plate in two halves and the lid in four quarters for the Bambu Lab A1. Every exported file was checked to be a single watertight body with no naked edges, which is how a corrupted fillet on part 03 was found. Figure 14 shows the base post sockets prepared in Bambu Studio.

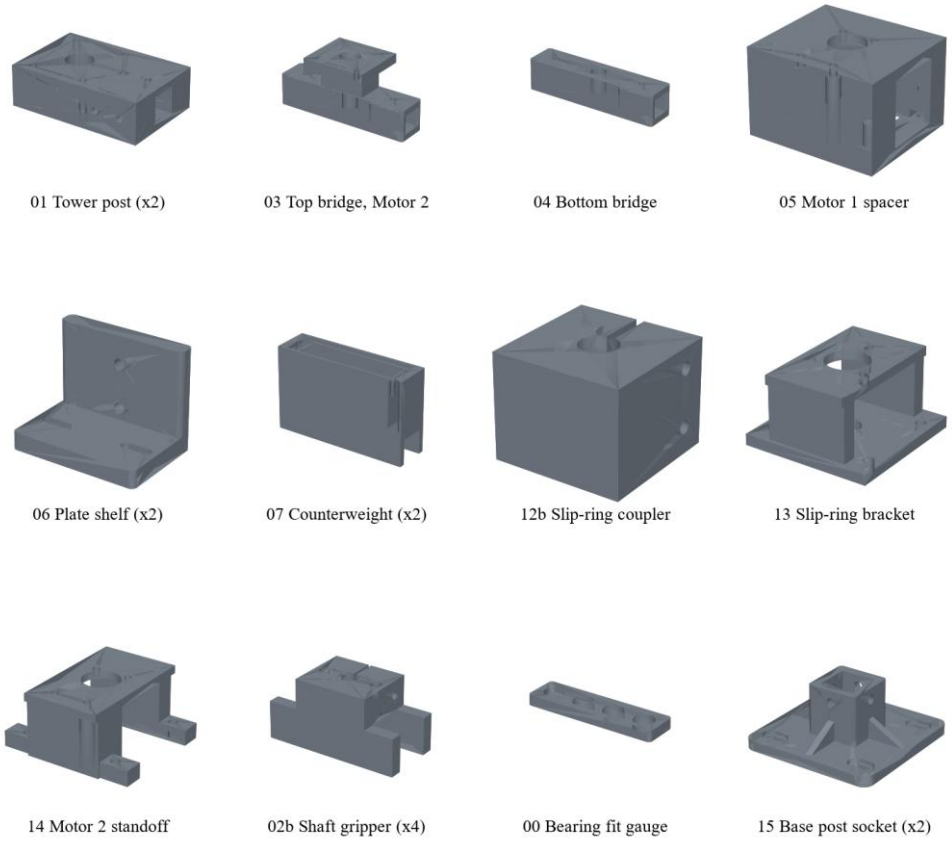


Figure 13 3D-printed parts of the rotocaster generated from the CadQuery model.

Table 12 Schedule of 3D-printed parts (mass at an effective PLA density of 744 kg/m³)

Part	Description	Qty	Size (mm)	Volume (cm³)	Mass each (g)
00	Bearing fit gauge (tool)	1	150 × 40 × 12	58.0	43
01	Tower post bearing holder	2	112 × 74 × 34	246.0	183
02/02b	Shaft gripper	4	72 × 30 × 38	46.6	35
03	Outer frame top bridge, Motor 2	1	150 × 62 × 68	196.1	146
04	Outer frame bottom bridge	1	150 × 34 × 34	125.6	94
05	Motor 1 spacer	1	70 × 62 × 46	108.2	80
06	Inner frame plate shelf	2	60 × 42 × 46	36.9	27
07	Counterweight (ballast pocket)	2	110 × 30 × 68	98.7	73
07b	Counterweight cover	2	95.6 × 25.6 × 3	7.1	5
08	Mould plate (2 halves)	1	180 × 300 × 8	200.9	150
10	Mould lid (4 quarters)	1	300 × 299.9 × 10	597.5	445
11	Mould pour plug	1	77 × 77 × 19	52.9	39
12b	Slip-ring coupler	1	25 × 25 × 20	10.6	8
13	Slip-ring bracket	1	70 × 72 × 41	63.4	47
14	Motor 2 coupler standoff	1	70 × 66 × 33	45.0	34
15	Base post socket	2	80 × 80 × 35	52.1	39
–	Control box body	1	166 × 96 × 63	149.0	111
–	Control box lid	1	166 × 96 × 5	71.6	53
	Total for one machine				**2,043**

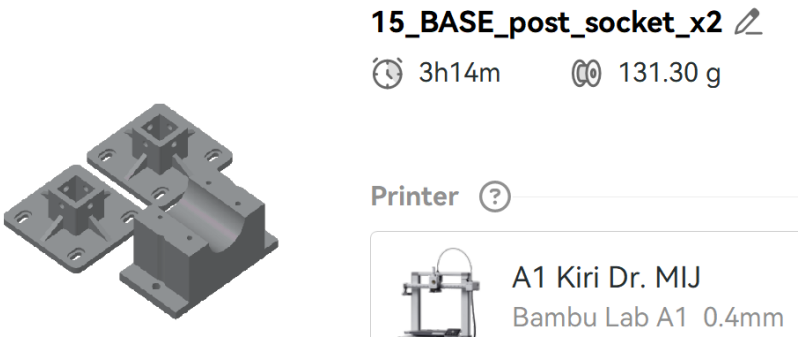


Figure 14 Base post sockets sliced in Bambu Studio for the Bambu Lab A1 (3 h 14 min, 131 g of PLA).

8.2 Bill of materials and cost

Components were sourced from online suppliers (Shopee); filament and printers were provided by the laboratory. The final cost of the mechanical build is about RM 464 (Table 13), against RM 367 planned at the design freeze. The difference comes mainly from structural parts and fasteners bought during assembly and from the third motor that replaced the one burnt out in the first powered test. The detailed list is in Appendix A.

Table 13 Budget of the rotocaster

Section	Main items	Cost (RM)
Structure	2020 aluminium extrusion (5 m), 12 gusset brackets, plywood base 750 × 450 × 18 mm	142
Fasteners	70 M5 drop-in T-nuts, M3–M6 screws, nuts and washers, M5 heat-set inserts, M8 ballast nuts	154
Rotating parts	10 × 608ZZ bearings, Ø8 mm chrome shaft, 2 rigid 6–8 mm couplers, 6-wire capsule slip ring	85
Motors	3 × JGY-370 4632 worm gearmotor, 12 V, 18 rpm	83
Total	Excluding electronics (Arduino, driver, display, adapter: about RM 50)	**464**

8.3 Assembly and fault log

The frame was assembled on its plywood base using a fastener schedule prepared for every printed part (Appendix B), giving the screw size, quantity, insertion direction and any part that must be fitted first. In total the frame uses 44 M5 drop-in T-nuts, 50 M5 screws of six lengths, 12 M4 screws and five 608ZZ bearings. Before assembly, design reviews and automated checks recorded 18 faults, and the first powered test added one more (Table 14). Six of them would have stopped the assembly, for example a bearing bore on the wrong axis and a mould rim 6 mm wider than the frame opening on each side. Every fault was traced to its cause and corrected in the CAD source.

Table 14 Summary of the fault log

Fault no.	Found by	Fault	Correction
1–6	Automated geometry audit	Bearing cap bore on the wrong axis; missing motor holes; post length put the bore 58 mm off-axis; M4 holes inside the M5 pattern; shelf screws on the wrong face; mould plate 340 mm wide for a 320 mm opening; cut list 34 mm short	Corrected in the source
7–10	Visual print review	Bridge hub in two disconnected pieces; slip-ring arm in two pieces; pour plug with no retention; redundant screw cover	Merged, redesigned, retention added, cover deleted
11–12	Interface review	Lid bolt circle inside the rim bore; mould rim overhanging the opening by 6 mm per side	Bolt circle moved to Ø280 mm; rim made flush at Ø300 mm
13–15	Fusion 360 checks	Counterweight with no rail to mount on; two hubs pointing inward through the rails	Split into two weights at x = ±120 mm; hubs reoriented
16–17	Motor specification check	Assumed hole patterns did not match the gearbox	18 × 33 mm pattern from the supplier drawing

Fault no.	Found by	Fault	Correction
18	Print and fit check	Motor 2 could not couple to the inner frame	Part 03 rebuilt with a second 608 bearing; new part 14 standoff
19	First powered test	Motor 1 stalled against an over-heavy counterweight and burnt out	Ballast reduced to near-neutral balance; motor replaced

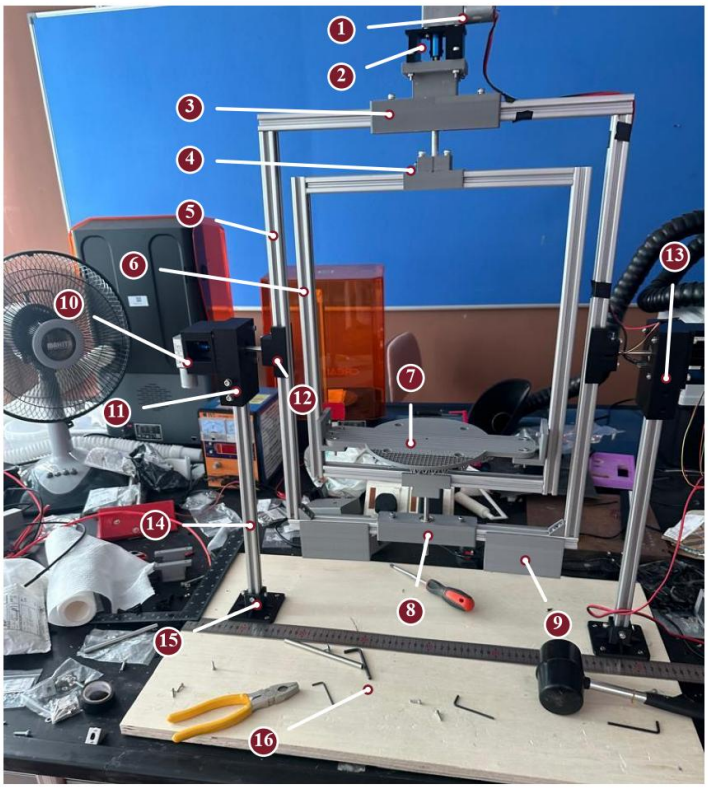
Other findings came from assembly. The 180 mm rails arrived at 184 mm, so the remaining rails were measured against the cut list. The M3 × 12 and M3 × 10 screws bottomed out in the motor's tapped holes before clamping, so shorter screws sized as the plate thickness plus the usable thread depth were used. The 2 × 60 mm shafts first bought were too short to reach the coupler, and all four shafts were standardised at 125 mm.

8.4 Commissioning

The electronics were commissioned in stages so that a wiring fault could not damage a motor or the frame: the sketch was uploaded with the motors unplugged; the 12 V line was checked for shorts and common ground; 12 V was applied with the motors disconnected to check the display and potentiometers; each motor was then run off the frame from a bench supply with a current limit of about 0.5 A; and finally the motors were fitted and run one axis at a time with the other locked and a hand on the RUN switch. This sequence is now Appendix C of the SOP.

8.5 First powered test and results

With the counterweights at the calculated mass, the frame came to rest with Motor 2 at the top, which showed that the real frame was counterweight-heavy: the as-built parts differ from the densities assumed in the CAD model. On every revolution, Motor 1 therefore had to lift the excess ballast over the top. It stalled, sparked and overheated within seconds, and would not run again even off the frame (fault 19). Nuts were then removed from the counterweights until the frame settled almost horizontal when released, with a residual torque comparable to bearing friction. A replacement motor was fitted, and both motors were then run from the controller with independent speed control from the two potentiometers, the RUN switch and the OLED display (Figure 15). Fine balancing with a real mould fitted remains to be completed.



Key

1 Motor 2, inner axis (JGY-370)
2 Motor 2 standoff and coupler (part 14)
3 Top bridge with Motor 2 pillar (part 03)
4 Inner-axis shaft gripper (part 02)
5 Outer frame, 2020 extrusion
6 Inner frame, 2020 extrusion
7 Mould plate (part 08) on plate shelves (part 06)
8 Bottom bridge (part 04)
9 Counterweights with M8 ballast (part 07, × 2)
10 Motor 1, outer axis, on its spacer (part 05)
11 Left tower bearing holder (part 01)
12 Outer-axis shaft gripper and trunnion (part 02)
13 Right tower with slip ring and Motor 2 wiring
14 Upright post, 2020 extrusion
15 Post socket (part 15)
16 Plywood base, 750 × 450 × 18 mm

Figure 15 The assembled dual-axis rotocaster with its main parts numbered.

The test produced practical operating rules, which were written into the SOP:

- A worm gearmotor stalls silently and burns out in seconds, so a new set-up must first be run on a current-limited supply.
- Balance must be checked before power is applied: the frame should hold its position at 90° when released.
- The heavy side is the one that ends up at the bottom at rest; weight is removed there or added opposite.
- The axis that is not being tested should be locked, otherwise its weight shifts and the balance readings change.
- The outer axis is trimmed at the counterweights: the counterweight screw is loosened, M8 nuts (about 5 g each) are added if Motor 2 swings down or removed if the counterweights swing down, keeping the same number on both sides, and the screw is then tightened so the nuts are clamped and cannot shift as the counterweight turns over. The inner axis is trimmed with the ±6 mm mould plate slots.
- Each real mould must be weighed and balanced with the frame, because a centred mould adds mass but almost no unbalance, while an off-centre one quickly uses up the motor margin.

9 Standard Operating Procedure

The SOP, document SOP-BFL-ROT-01, expands the short operating procedure written at the end of the build into a complete document. It covers purpose and scope, responsibilities, a labelled description of the machine, its controls, specifications, operating limits and mould interface, a hazard and control table with PPE, and step-by-step procedures with acceptance checks for pre-use inspection, mounting and static balancing, filling, start-up and running, shutdown and demoulding, and cleaning, with a dedicated step table for adjusting the M8-nut ballast in the counterweights. It

also contains emergency actions, a maintenance schedule, a troubleshooting guide, a pre-use and balance checklist, a run log, a training sign-off, the load chart and the commissioning procedure.

The visual SOP (Figure 16) is a one-page colour poster for display next to the machine. It shows the PPE and hazards, the five phases of a run (prepare, balance, fill, run, shut down), the do's and don'ts, the key limits and the emergency actions.



Figure 16 Visual SOP poster for the dual-axis rotating device (full size in SOP-BFL-ROT-01, Appendix A).

10 Discussion

10.1 PDS compliance

Table 15 Compliance with the product design specification

ID	Requirement	Status	Evidence or comment
a	Dual-axis, 360° across orthogonal axes	Met	Two perpendicular axes, 360° continuous each, independently driven by two motor-driven shafts
b	Smooth operation up to 100 RPM	Deviation (justified)	Smooth 15 s soft start; top speed 18 rpm no-load. 100 rpm is not achievable with worm gearmotors at this scale and is above the 77 rpm centrifugal limit. Recommend revising to 0–20 rpm with a load chart

ID	Requirement	Status	Evidence or comment
c	Precise speed control mechanisms	Partly met	One potentiometer per axis (OUTER, INNER), 8-sample averaging, PWM with a 15 s ramp; settings are repeatable but open-loop. Add an encoder and closed-loop speed control for precise speed
d	Wall-socketted power supply	Met	12 V DC wall adapter with a 2 A fuse
e	ON/OFF via operator panel	Met	RUN/STOP rocker switch on the control box lid; unplugging the adapter isolates the machine
f	LED display of RPM and status	Partly met	OLED shows RUN/STOP, duty and an estimated rpm for each axis. Fit an encoder or Hall sensor for a measured speed
g	3D-printed or equivalent materials	Met	About 2 kg of PLA parts generated from the CAD model; materials selection in Section 10.3
h	Quick-release or minimal-effort mould mounting	Partly met	Standard bolt-on interface: mould foot 4 × M6 on Ø140 mm PCD, plate slotted for balancing. Minimal effort but not quick-release; captive thumb screws or a quarter-turn clamp on the plate would make it tool-free
i	Mould Ø300 × 300 mm, 2–10 kg	Partly met	Envelope met; 5 kg limit with the fitted motors, 10 kg with a 70 kg·cm motor
j	No cable wind-up; no dripping (objective)	Met by design	Slip ring through a hollow trunnion; sealed lid and retained pour plug. Confirm in the first casting trial

10.2 Documentation, quality control and safety

Table 16 shows how the professional notes of the brief on documentation, quality control and safety were met.

Table 16 How the professional notes of the brief were met

Professional note	Measures taken
Detailed technical documentation for all tasks	Parametric CadQuery source that regenerates every part and STL file This report with calculations, bill of materials, fastener schedule, fault log and control sketch SOP-BFL-ROT-01 (written) with a one-page visual SOP poster
Quality control	Bearing fit gauge printed first; bore set from the gauge result Automated geometry audit of the CAD model (faults 1–6) and watertight check of every exported STL Fusion 360 motion check of both axes through 360° Every delivered part checked against the design (184 mm rails, motor hole pattern, screw lengths) Fault log of 19 faults, each corrected in the source model rather than by hand Staged commissioning with a current-limited supply; pre-use and balance checklists in the SOP
Safety protocols	Self-locking worm gears hold the frames when stopped; 2 A fuse; immediate stop on RUN/STOP Load chart and 5 kg limit; balance before power Hazard and control table, PPE, keep-out zone, emergency actions and lock-out by unplugging in the SOP Component margins checked: bearings 45 ×, PLA D-flat 16 ×, shaft 21 ×, slip ring 5.7 ×

10.3 Materials selection

Materials were chosen for their mechanical properties and for how easily they could be made or bought in the laboratory (Table 17).

Table 17 Materials selection

Part	Material	Mechanical reason	Manufacturing reason
Frames, towers	Aluminium 2020 T-slot extrusion	Stiff and light (2700 kg/m³); keeps rotating mass and inertia low	Cut to length; parts clamp with T-nuts, so positions can be adjusted without machining

Part	Material	Mechanical reason	Manufacturing reason
Brackets, bridges, grippers, bearing housings, mould plate and lid, control box	PLA, 0.2 mm layers, 40 % infill, 6 walls at bearing seats	Yield about 45 MPa against 2.8 MPa peak stress at the D-flats (16 ×); accurate, low-warp prints	Printed on the laboratory's printers from the parametric model; large parts split to fit the bed
Shafts	Hardened chrome steel, Ø8 mm	Yield about 250 MPa against 12 MPa bending (21 ×); hard surface for bearings	Standard linear shaft; D-flats filed by hand; cut to 125 mm
Bearings	608ZZ deep-groove ball bearings	3400 N dynamic rating against 75 N load (45 ×); sealed against silicone	Low-cost standard part; press-fit into printed 22.2 mm bores
Base	18 mm plywood	Stiff and heavy enough to resist tipping	Easy to cut, drill and fix with wood screws
Counterweight ballast	Zinc-plated steel M8 nuts	Dense (7850 kg/m³), so little volume is needed	Cheap; adjustable in small steps by adding or removing nuts
Hollow right trunnion	Steel tube Ø8 mm, 3 mm bore	Carries the trunnion load while passing the slip-ring wires	Standard tube; held by a clamping gripper instead of a D-flat

NOTE PLA softens at about 60 °C. It is suitable for room-temperature-cure silicones. If a heat-cured silicone or an oven cure on the machine is planned, the printed parts should be reprinted in PETG or ABS.

10.4 Market readiness

Table 18 Market-readiness assessment

Aspect	Status	Action
Function	Both axes run under independent speed control	Balance and cast the first real mould; record the result
Mould mounting	Bolt-on interface, minimal effort	Add captive thumb screws or a quarter-turn clamp for tool-free quick release (PDS h)
Drive capacity	Fitted motors limit moulds to 5 kg	Fit 16 rpm, 70 kg·cm motors for moulds up to 10 kg
Motor protection	Single 2 A fuse; L298N has no current limit	Add a fuse or current limit per motor, or a driver with current sensing and stall detection
Speed control and indication	Open-loop; speed estimated from duty cycle	Fit an encoder or Hall sensor on each axis; close the speed loop and display the measured rpm (PDS c, f)
Guarding	Open frame; keep-out zone by procedure	Add a clear enclosure or light guard that stops the machine when opened
Isolation	Unplug the adapter	Add a panel power switch on the 12 V input
Labelling	Lid engraved with title and control names	Add a rating label, a rotating-parts warning and the SOP number
Documentation	Written SOP, visual SOP, this report, CAD source	Archive CAD, STL, firmware and documents together; review after first use

10.5 Lessons learned

- A requirement such as 100 rpm must be checked against both the drive and the physics of the process before it is accepted.
- A design that is correct on screen can still fail because a supplier's part differs from its listing, a printed hole is undersize or the real frame is balanced differently from the model. Checking every delivered part and printing a fit gauge first saved rework.
- Correcting the parametric source instead of patching parts by hand kept the CAD and the hardware in step through 19 faults.
- New drives must be tested with current limiting and a balanced load, one axis at a time.

11 Conclusion and Recommendations

A dual-axis rotocaster for silicone moulds of up to Ø300 × 300 mm was designed from a parametric CAD model, analysed, built for about RM 464 plus about RM 50 of electronics, balanced and run under an Arduino-based controller in a purpose-designed box. The analysis showed that the machine is unbalance-limited, that a 409 g counterweight is needed for the inner-axis motor, and that the 100 rpm target should be replaced by a load chart at 10–20 rpm. Comprehensive technical documentation and a detailed written and visual SOP were prepared, completing the three tasks of the brief.

The following actions are recommended:

1. Weigh and balance the first real mould, cast a part and record the wall thickness at several points.
2. Keep moulds at 5 kg or less, or fit 16 rpm, 70 kg·cm worm gearmotors for heavier moulds.
3. Add current limiting or a fuse for each motor, and a panel power switch.
4. Fit an encoder or Hall sensor on each axis, close the speed loop and display the measured speed, to meet PDS items c and f fully.
5. Add captive thumb screws or a quarter-turn clamp to the mould plate so moulds can be fitted without tools (PDS item h).
6. Add a guard around the frame and rating and warning labels.
7. Revise the PDS speed item to 0–20 rpm with the load chart in SOP-BFL-ROT-01 Appendix B.

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Appendix A Detailed Bill of Materials

Table 19 Detailed bill of materials

Section	Item	Specification	Qty	Cost (RM)
Structure	2020 aluminium extrusion	6 mm slot, cut to the list in Table 4	5 m	75
	Gusset corner bracket	20-series, RM 1.66 each	12	20
	Plywood base	750 × 450 × 18 mm	1	26
	Structure subtotal	Including parts bought during assembly		**142**
Fasteners	M5 drop-in T-nut	For the 2020 slot	70	49
	Screws	M5 × 8, M5 × 16, M6 × 25, M4 × 12, M3 × 10	Mixed	17
	Hex nuts	M3, M4, M5, M6	60	10
	Washers	M5, M6	40	6
	Heat-set inserts	M5	8	12

Section	Item	Specification	Qty	Cost (RM)
	M8 nuts, plain zinc	Counterweight ballast	About 70	10
	Fasteners subtotal	Including parts bought during assembly		**154**
Rotating parts	Ball bearing 608ZZ	8 × 22 × 7 mm, RM 0.66 each	10	
	Linear shaft	Ø8 mm hardened chrome, D-flats filed by hand	1	
	Rigid coupling	6 mm to 8 mm	2	
	Capsule slip ring	6-wire, 22 mm, 2 A per circuit	1	
	Rotating parts subtotal			**85**
Motors	JGY-370 4632 worm gearmotor	12 V, 18 rpm, 13 kg·cm, self-locking; third motor replaced the burnt Motor 1	3	83
Total	Mechanical build	Excluding electronics		**464**
Electronics	Controller set	Arduino Uno, L298N, 1.3-inch I2C OLED, 2 potentiometers, ON-OFF-ON rocker switch, 12 V adapter, 2 A fuse, capacitors	1 set	About 50

Appendix B Fastener Schedule

Table 20 Fastener schedule of the rotocaster

Part	Fasteners	Notes
Gusset brackets (8 corners)	16 × M5 × 8, 16 × M5 T-nut	Two per gusset, one into each rail
01 Tower post (× 2)	4 × M5 × 35, 4 × M5 × 16, 8 × M5 T-nut, 2 × 608ZZ	Side clamp screws enter from the inboard face so the heads clear part 13; bearings press-fit from outboard
02/02b Shaft gripper (× 4)	8 × M5 × 25, 8 × M5 T-nut; 02b adds M4 × 40 and M4 nut	02b pinch screw closes the split bore on the hollow right trunnion
03 Top bridge, Motor 2	2 × M5 × 16, 2 × M5 T-nut, 2 × 608ZZ, 4 × M4 × 25, 4 × M4 nut	M4 screws also carry part 14
04 Bottom bridge	2 × M5 × 16, 2 × M5 T-nut, 1 × 608ZZ	Bearing from underneath
05 Motor 1 spacer	4 × M5 × 45, 4 × M5 nut, 4 × M3	Nuts trapped in hex pockets; M3 length = 6 mm plate + usable thread depth
06 Plate shelf (× 2)	4 × M5 × 16, 4 × M5 T-nut, 4 × M4 × 25, 4 × M4 nut	M4 screws in the balance slots are only nipped until balanced
07 Counterweight (× 2)	4 × M5 × 55, 4 × M5 T-nut	At x = ±120 mm; ballast as found by balancing
07b Counterweight cover (× 2)	8 × M3 × 12	Self-tapping; load the ballast first
08 Mould plate	4 × M6, 8 × M6 washer	Ø140 mm PCD; carries the whole mould
10 Mould lid	6 × M5, 12 × M5 washer; 3 × M4 × 20, 3 × M4 nut	Ø280 mm PCD, star pattern; pour plug with O-ring
12b Slip-ring coupler	2 × M3 × 12	Pinch screws on the trunnion tube and rotor stub
13 Slip-ring bracket	2 × M5 × 45, 2 × M5 nut and washer, 3 × M3 × 12, 3 × M3 nut	Slotted flange holes let the ring rotate to line up
14 Motor 2 standoff	4 × M3 (about M3 × 8), 4 × M4 × 25 shared with part 03	Drive the motor screws up from inside before fitting to part 03

Appendix C Control Sketch

Arduino sketch rotocaster_controller.ino (requires the U8g2 library by olikraus).

```
// Rotocaster controller: 2 pots -> 2 worm gearmotors, 1.30" 128x64 I2C OLED
// UTM / Biofluid Lab, Dr. Zahran
//
// Hardware: Arduino Uno, L298N driver, SH1106 OLED (SDA=A4, SCL=A5), RUN switch.
// Library:  "U8g2" by olikraus (Tools > Manage Libraries).
// L298N:    remove the ENA and ENB jumpers. Speed = PWM on EN pins, direction = IN pins.
// Switch:   ON-OFF-ON rocker. Middle pin to GND, one outer pin to D2 = RUN.
//
// Memory: the OLED uses the PAGE buffer (_1_), 128 bytes, instead of the full
// buffer (_F_), 1024 bytes. That keeps the Uno's 2 KB of RAM mostly free.
```

```
#include <Wire.h>
#include <U8g2lib.h>

// Most 1.30" modules are SH1106. If the picture is shifted right by 2 pixels or
// has a noisy strip at the edge, swap to the SSD1306 line.
U8G2_SH1106_128X64_NONAME_1_HW_I2C u8g2(U8G2_R0, U8X8_PIN_NONE);
// U8G2_SSD1306_128X64_NONAME_1_HW_I2C u8g2(U8G2_R0, U8X8_PIN_NONE);

// Pins
const uint8_t POT_OUTER = A0, POT_INNER = A1;
const uint8_t M1_EN = 5, M1_IN1 = 7, M1_IN2 = 8;      // outer axis: ENA, IN1, IN2
const uint8_t M2_EN = 6, M2_IN1 = 9, M2_IN2 = 10;     // inner axis: ENB, IN3, IN4
const uint8_t RUN_SW = 2;                             // to GND = RUN, open = STOP

// Tuning
const float MAX_RPM   = 18.0;    // nameplate no-load. Display is an estimate (no encoder)
const int   MIN_PWM   = 60;      // lowest PWM that reliably starts a motor. L298N drops ~2 V,
                                 // so this is higher than on a bare driver. Tune on the bench.
const int   MAX_PWM   = 255;
const int   POT_DEAD  = 30;      // pot counts near zero = off
const float RAMP_TIME = 15.0;    // seconds from 0 to full, as in the torque calculation
const unsigned long LOOP_MS = 20;
const unsigned long LCD_MS  = 300;

float outPwm[2] = {0, 0};        // current (ramped) PWM per motor
unsigned long lastLoop = 0, lastLcd = 0;

int readPot(uint8_t pin) {       // 8-sample average to stop jitter
  long s = 0;
  for (uint8_t i = 0; i < 8; i++) s += analogRead(pin);
  return s / 8;
}

int potToPwm(int raw) {
  if (raw < POT_DEAD) return 0;
  return map(raw, POT_DEAD, 1023, MIN_PWM, MAX_PWM);
}

void drive(uint8_t en, uint8_t in1, uint8_t in2, int pwm) {
  digitalWrite(in1, HIGH);       // one direction only; swap HIGH/LOW (or the motor wires) to reverse
  digitalWrite(in2, LOW);
  analogWrite(en, pwm);          // 0 = stopped (worm gear self-locks)
}

void setup() {
  pinMode(RUN_SW, INPUT_PULLUP);
  pinMode(M1_EN, OUTPUT); pinMode(M1_IN1, OUTPUT); pinMode(M1_IN2, OUTPUT);
  pinMode(M2_EN, OUTPUT); pinMode(M2_IN1, OUTPUT); pinMode(M2_IN2, OUTPUT);
  drive(M1_EN, M1_IN1, M1_IN2, 0);
  drive(M2_EN, M2_IN1, M2_IN2, 0);

  u8g2.begin();
  u8g2.setBusClock(400000);      // faster I2C, shorter screen refresh
  u8g2.setFont(u8g2_font_7x13B_tr);
  u8g2.firstPage();
  do {
    u8g2.drawStr(18, 28, "ROTOCASTER");
    u8g2.drawStr(30, 46, "CONTROLLER");
  } while (u8g2.nextPage());
  delay(1200);
}

// Builds "Outer  72%  9.4rpm" into buf
void formatLine(char *buf, size_t n, const char *name, float pwm) {
  int pct = (int)(pwm * 100.0 / MAX_PWM + 0.5);
  float rpm = pwm * MAX_RPM / MAX_PWM;
  char r[8];
  dtostrf(rpm, 4, 1, r);
  snprintf(buf, n, "%s %3d%% %srpm", name, pct, r);
}

void drawScreen(bool run) {
  char l1[24], l2[24];
  formatLine(l1, sizeof(l1), "Outer", outPwm[0]);
  formatLine(l2, sizeof(l2), "Inner", outPwm[1]);

  u8g2.firstPage();
  do {
    u8g2.setFont(u8g2_font_7x13B_tr);
    u8g2.drawStr(0, 12, run ? "RUN" : "STOP  (switch off)");
    u8g2.drawHLine(0, 16, 128);
    u8g2.setFont(u8g2_font_6x12_tr);
    u8g2.drawStr(0, 36, l1);
    u8g2.drawStr(0, 56, l2);
  } while (u8g2.nextPage());
}
```

```
void loop() {
  unsigned long now = millis();
  if (now - lastLoop < LOOP_MS) return;
  float dt = (now - lastLoop) / 1000.0;   // real elapsed time, so the ramp stays 15 s
  lastLoop = now;                          // even when the screen redraw takes a while

  bool run = (digitalRead(RUN_SW) == LOW);
  int target[2] = {0, 0};
  if (run) {
    target[0] = potToPwm(readPot(POT_OUTER));
    target[1] = potToPwm(readPot(POT_INNER));
  }

  // Ramp toward target. Stopping is immediate (switch off or pot at zero).
  const float step = (float)MAX_PWM / RAMP_TIME * dt;
  for (uint8_t i = 0; i < 2; i++) {
    if (target[i] == 0)                outPwm[i] = 0;
    else if (outPwm[i] < target[i])    outPwm[i] = min(outPwm[i] + step, (float)target[i]);
    else                               outPwm[i] = max(outPwm[i] - step, (float)target[i]);
    // A running motor never sits below MIN_PWM, so it can't stall in the low range.
    if (target[i] > 0 && outPwm[i] < MIN_PWM) outPwm[i] = MIN_PWM;
  }

  drive(M1_EN, M1_IN1, M1_IN2, (int)outPwm[0]);
  drive(M2_EN, M2_IN1, M2_IN2, (int)outPwm[1]);

  if (now - lastLcd > LCD_MS) {
    lastLcd = now;
    drawScreen(run);                     // blocks ~50 ms; motors keep their last PWM
  }
}
```